

Methunraj A

Bachelor of Technology in Computer Science and Engineering
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LinkedIn Profile

Github Profile

EDUCATION

- Bachelor of Technology – Computer Science and Engineering (Honors: Quantum Computing)** 2023 – 2027
SRM Institute of Science and Technology, Tiruchirappalli CGPA: 9.50
- Higher Secondary Certificate (12th Grade)** 2022 – 2023
Bharani Park Matric Hr Sec School, Karur Percentage: 82.6%
- Secondary School Certificate (10th Grade)** 2020 – 2021
Gurudevvar Matric Hr Sec School, Karur Percentage: 81.8%

AWARDS & ACHIEVEMENTS

- Apple Student Swift Challenge 2026 – Winner** March 2026
Selected as a winner of Apple's globally competitive Swift Student Challenge 2026; developed an interactive Swift Playgrounds app demonstrating innovative use of SwiftUI and Apple frameworks, recognized by Apple among thousands of global student submissions.

EXPERIENCE

- Infosys Virtual Internship 6.0** Aug 2025 – Oct 2025
Infosys Springboard – Artificial Intelligence & NLP Remote
 - Developed an AI-powered application to analyze and quantify skill gaps between job descriptions and candidate resumes using Python and Natural Language Processing (NLP).
 - Built a document processing pipeline supporting PDF, DOCX, and TXT formats; applied Named Entity Recognition (NER) with spaCy for accurate skill extraction.
 - Integrated Sentence-BERT for semantic similarity analysis, enabling intelligent skill comparison beyond keyword matching to improve resume-to-job-description alignment.
- Defence Research & Development Organisation (DRDO), CVRDE, Avadi** Jun 2025 – Jul 2025
Research Intern – Computer-Aided Manufacturing Onsite
 - Project: **Development of Computer-Aided Process Planning (CAPP) for Objects with Bézier Curve Profile.**
 - Developed optimized CNC toolpaths using quintic Bézier curves; performed control point extraction, curve reconstruction, and Bernstein polynomial analysis for precision manufacturing.
- AICTE Virtual Internship – AI/ML** Jul 2024 – Sep 2024
All India Council for Technical Education (AICTE) Remote
 - Completed hands-on training in supervised and unsupervised machine learning, model evaluation metrics, and implemented end-to-end ML projects using Python and Scikit-learn.

PROJECTS

- The Last Spark – iOS Arcade Game (Apple Swift Student Challenge Winner)** SwiftUI
 - Built a sci-fi arcade dodge game using SwiftUI with custom game loop, scoring system, and dynamic difficulty.
 - Implemented story and endless modes with haptics, audio, and smooth UI transitions for immersive gameplay.
- FedCausalFreshNet – AI Food Spoilage Prediction System** Python, FastAPI, PyTorch
 - Developed a smart system to predict food shelf-life using environmental data and storage conditions.
 - Applied federated learning and causal inference (PyTorch, DoWhy, Flower) for privacy-preserving recommendations.
- Lost and Found Management System** React, Flask, MySQL
 - Built a full-stack Lost & Found web app with auth + image upload similarity matching.
 - Tech: React (Vite+TS), Tailwind, Shadcn UI, Flask REST API, MySQL.
- Air Quality Prediction System** Python, XGBoost, Flask
 - Built a full-stack system to predict hyper-local PM2.5 levels in Delhi using real-time API data.
 - Trained an XGBoost model with geo-features and deployed via Flask API + Leaflet.js heatmap.

RESEARCH & PUBLICATIONS

- Green Task: A Carbon-Aware Scheduling Algorithm for Enhancing Energy Efficiency in Edge-Fog Computing Systems**
International Journal for Science and Advanced Technology (IJSAT), Vol. 16, Issue 3, 2025.
Co-authored paper proposing a carbon-aware task scheduling algorithm for edge-fog environments; led algorithmic complexity analysis and simulation-based performance evaluation.

TECHNICAL SKILLS

Programming Languages: Python, C, C++, SQL, Swift, HTML, CSS

Frameworks & Libraries: SwiftUI, NumPy, Pandas, Matplotlib, Seaborn, Scikit-learn, Flask, Sentence-BERT, spaCy

Tools & Platforms: Git, GitHub, Snowflake, Jupyter Notebook, VS Code, Xcode, Swift Playground

Areas of Expertise: Machine Learning, Natural Language Processing (NLP), Data Engineering, Software Engineering, Cloud & AI Integration, iOS App Development

Soft Skills: Problem Solving, Analytical Thinking, Self-Learning, Adaptability, Team Collaboration

CERTIFICATIONS

- **SnowPro Associate: Platform** – Snowflake
- **Introduction to Computer Vision and Image Processing** – IBM
- **Machine Learning** – Internshala
- **Joy of Computing using Python** – NPTEL
- **Database Management System** – NPTEL